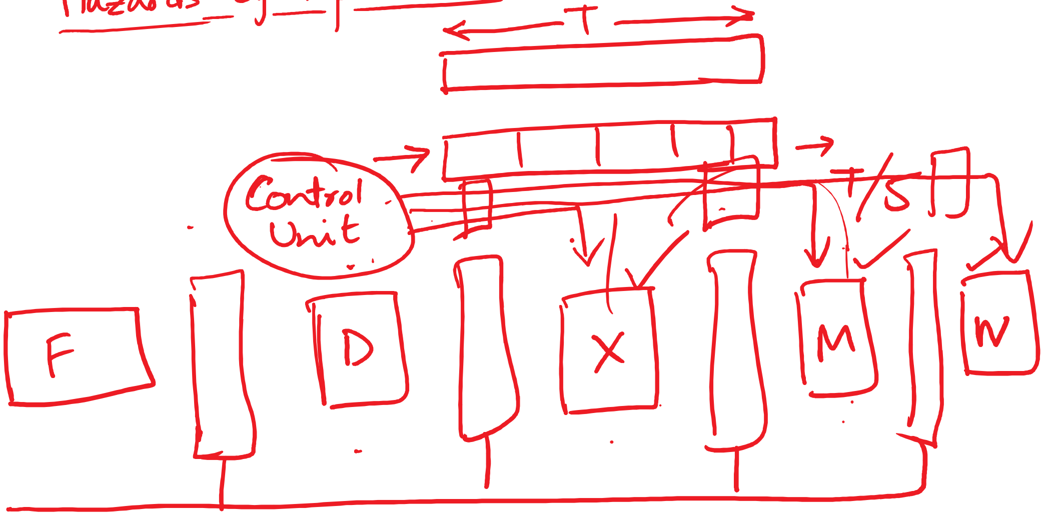
Lec 20

Thursday, July 7, 2022 11:25 AM

Hazards of Pipeline



How do we analyze performance of processors?

"Iron law" of processor performance

$$\frac{\text{Time}}{\text{Program}} = \frac{\text{Instructions}}{\text{Program}} \times \underbrace{\frac{\text{Cycles}}{\text{Instruction}}}_{\text{depends on } \mu\text{architecture}} \times \frac{\text{Time}}{\text{Cycle}}$$

Depends on ISA,

Technology

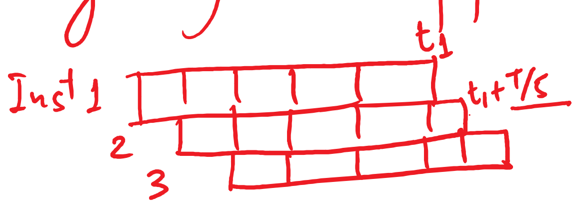
CPI = Cycles per Inst^n.

IPC = Inst^n " Cycle $= \frac{1}{CPI}$

T sec

Single-cycle un-pipelined : CPI = 1.

cycle time: long



Pipelined — · CPI = 1, cycle time: short

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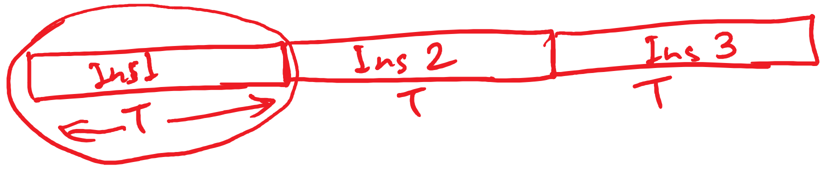

Unpipelined

$$\frac{\text{Time}}{\text{Prog.}} = \frac{3}{1} \times \frac{1}{1} \times \frac{T}{1}$$
$$= \underline{3T}$$

Pipelined:

$$\frac{\text{Time}}{\text{Prog.}} = \frac{3}{1} \times \frac{1}{1} \times \frac{T}{5}$$
$$= \frac{3T}{5}$$

	1	2	3	4	5	6	
Ins1	F	D	X	M	W		
2 →		F	D	X	M	W	
3			F	D	X	M	W
4							
5							

ADD R1, R2, R3
ADD R4, R1, R5

DATA Hazard:

RAW → Read after Write

	1	2	3	4	5	6	7	8	9	10
ADD	F	D	X	M	W					
ADD		F	D	D	D	D	X	M	W	
Ins			F	F	F	F	D	X	M	W

Data Hazard: occurs when one Instn depends on a data value produced by a preceding Instr still in the pipeline.